

Homework 6

ORF 245 - Fundamentals of Statistics

Due: Apr. 4th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
 - Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).
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Exercise: 1. (Bias-Variance Decomposition)

Let $\hat{\theta}(X_1, \dots, X_n)$ be any point estimator for some unknown population parameter θ . Recall that the Mean Squared Error (MSE) for an estimator $\hat{\theta}(\cdot)$ is defined as:

$$\text{MSE}(\hat{\theta}; \theta) = \mathbb{E} \left[(\hat{\theta} - \theta)^2 \right].$$

Show that $\text{MSE}(\hat{\theta}; \theta) = \text{Bias}(\hat{\theta}; \theta)^2 + \text{Var}(\hat{\theta})$.

- **Solution:** To prove the bias-variance decomposition, note that we can write

$$\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta).$$

Squaring both sides gives

$$(\hat{\theta} - \theta)^2 = (\hat{\theta} - \mathbb{E}[\hat{\theta}])^2 + 2(\hat{\theta} - \mathbb{E}[\hat{\theta}])(\mathbb{E}[\hat{\theta}] - \theta) + (\mathbb{E}[\hat{\theta}] - \theta)^2.$$

Taking expectations and noting that

$$\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])] = 0,$$

we obtain

$$\mathbb{E} \left[(\hat{\theta} - \theta)^2 \right] = \text{Var}(\hat{\theta}) + \left(\mathbb{E}[\hat{\theta}] - \theta \right)^2.$$

This completes the proof.

Exercise: 2. (MLE of Poisson)

Let the discrete random variable X have a Poisson distribution with unknown parameter $\lambda > 0$. The pmf of X is given by:

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

Recall that a Poisson random variable appropriately models the number of arrivals within a given unit of time, and that its mean is given by λ .

- (a) If we observe 8 arrivals in one unit of time, compute the likelihood and log-likelihood functions. (I.e. we obtain a single realization of X given by $x = 8$.)

• **Solution:** Given a single observation $x = 8$, the likelihood function is

$$\mathcal{L}(\lambda; x = 8) = \frac{e^{-\lambda} \lambda^8}{8!}.$$

The log-likelihood function is then

$$\ell(\lambda; 8) = \log \mathcal{L}(\lambda; 8) = -\lambda + 8 \log(\lambda) - \log(8!).$$

- (b) Find the maximum likelihood estimate of λ given the setting in part (a), where we observe 8 arrivals, i.e. find $\hat{\lambda}_{MLE}(8)$.

• **Solution:** To find the MLE, differentiate the log-likelihood with respect to λ :

$$\frac{d}{d\lambda} \ell(\lambda; 8) = -1 + \frac{8}{\lambda}.$$

Setting the derivative equal to zero,

$$-1 + \frac{8}{\lambda} = 0 \quad \implies \quad \lambda = 8.$$

Thus, $\hat{\lambda}_{MLE}(8) = 8$.

- (c) Now, suppose that we have $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poi}(\lambda)$. Find the likelihood and log-likelihood functions.

• **Solution:** For $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\lambda)$, the likelihood function is

$$\mathcal{L}(\lambda; X_1, \dots, X_n) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i} \prod_{i=1}^n \frac{1}{x_i!}.$$

The log-likelihood function is

$$\ell(\lambda; X_1, \dots, X_n) = -n\lambda + \left(\sum_{i=1}^n x_i \right) \log(\lambda) - \sum_{i=1}^n \log(x_i!).$$

(d) Find the maximum likelihood estimator $\hat{\lambda}_{MLE}(X_1, \dots, X_n)$ for λ .

• **Solution:** Differentiating the log-likelihood with respect to λ :

$$\frac{d}{d\lambda} \ell(\lambda; X_1, \dots, X_n) = -n + \frac{1}{\lambda} \sum_{i=1}^n x_i.$$

Setting this derivative equal to zero gives

$$-n + \frac{1}{\lambda} \sum_{i=1}^n x_i = 0 \implies \lambda = \frac{1}{n} \sum_{i=1}^n x_i.$$

Hence, the MLE is

$$\hat{\lambda}_{MLE}(X_1, \dots, X_n) = \bar{X}.$$

(e) Is the estimator $\hat{\lambda}_{MLE}(X_1, \dots, X_n)$ biased? Justify your answer.

• **Solution:** Each X_i follows a Poisson distribution with mean λ , so $\mathbb{E}[X_i] = \lambda$. Hence, the sample mean satisfies

$$\mathbb{E}[\bar{X}] = \lambda,$$

which shows that $\hat{\lambda}_{MLE}(X_1, \dots, X_n)$ is an unbiased estimator of λ .

Exercise: 3. (MLE of a New Distribution)

Let $X_1, \dots, X_n$ be i.i.d. random variables, with each of their marginal pdfs given by:

$$f(x; \nu) = \begin{cases} 0, & \text{if } x < 0 \\ \nu^{-x} \ln(\nu), & \text{if } x \geq 0, \end{cases} \quad (1)$$

where $\nu > 1$ is an unknown parameter.

(a) Compute the likelihood function $\mathcal{L}(\nu; X_1, \dots, X_n)$.

• **Solution:** For $X_1, \dots, X_n$ with pdf as given, the likelihood function is

$$\mathcal{L}(\nu; X_1, \dots, X_n) = \prod_{i=1}^n [\nu^{-x_i} \ln(\nu)] = [\ln(\nu)]^n \nu^{-\sum_{i=1}^n x_i},$$

provided that $X_i \geq 0$ for all i .

(b) Find the maximum likelihood estimator $\hat{\nu}_{MLE}(X_1, \dots, X_n)$.

- **Solution:** The log-likelihood function is

$$\ell(\nu; X_1, \dots, X_n) = n \ln(\ln(\nu)) - \left(\sum_{i=1}^n x_i \right) \ln(\nu).$$

Differentiating with respect to ν gives

$$\frac{d}{d\nu} \ell(\nu) = \frac{n}{\nu \ln(\nu)} - \frac{\sum_{i=1}^n x_i}{\nu}.$$

Set the derivative equal to zero:

$$\frac{n}{\nu \ln(\nu)} - \frac{\sum_{i=1}^n x_i}{\nu} = 0 \quad \implies \quad \frac{n}{\ln(\nu)} = \sum_{i=1}^n x_i.$$

Solving for ν ,

$$\ln(\nu) = \frac{n}{\sum_{i=1}^n x_i} \quad \implies \quad \hat{\nu}_{MLE}(X_1, \dots, X_n) = \exp \left(\frac{n}{\sum_{i=1}^n x_i} \right).$$

Exercise: 4. (MLE for Variance of a Normally Distributed Population)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \theta)$ be i.i.d. random variables from a Normal distribution with mean μ and variance θ .

In this problem, **assume the population mean μ is known**, and the population variance θ is unknown.

- (a) Compute the likelihood function $\mathcal{L}(\theta; X_1, \dots, X_n)$.

- **Solution:** With μ known, the likelihood function is

$$\mathcal{L}(\theta; X_1, \dots, X_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta}} \exp \left(-\frac{(X_i - \mu)^2}{2\theta} \right) = \frac{1}{(2\pi\theta)^{n/2}} \exp \left(-\frac{1}{2\theta} \sum_{i=1}^n (X_i - \mu)^2 \right).$$

- (b) Compute the maximum likelihood estimator $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ for the population variance θ .

- **Solution:** The log-likelihood function is

$$\ell(\theta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\theta) - \frac{1}{2\theta} \sum_{i=1}^n (X_i - \mu)^2.$$

Differentiating with respect to θ :

$$\frac{d}{d\theta} \ell(\theta) = -\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{i=1}^n (X_i - \mu)^2.$$

Setting the derivative to zero,

$$-\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{i=1}^n (X_i - \mu)^2 = 0 \implies \theta = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2.$$

Thus,

$$\hat{\theta}_{MLE}(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2.$$

(c) Is $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ an unbiased estimator? Justify your answer. (A proof is not necessary, but certainly acceptable.)

• **Solution:** Since μ is known, for each i we have $\mathbb{E}[(X_i - \mu)^2] = \theta$. Therefore,

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] = \theta.$$

This shows that $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ is unbiased.

Exercise: 5. [OPTIONAL] (Point Estimation for Uniform RV's)

Note: This is an *optional* exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Let $X_1, \dots, X_n$ be i.i.d. random variables according to a uniform distribution on the interval $[0, \theta]$, where $\theta > 0$ is an unknown parameter. Each of their marginal probability density function is given by:

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} & \text{if } x \in [0, \theta] \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $\hat{\theta}_{AVG}(X_1, \dots, X_n) = 2\bar{X}_n$ is an unbiased estimator of θ .

• **Solution:** For $X_i \sim \text{Uniform}(0, \theta)$, the expected value is

$$\mathbb{E}[X_i] = \frac{\theta}{2}.$$

Thus,

$$\mathbb{E}[2\bar{X}_n] = 2 \left(\frac{\theta}{2} \right) = \theta,$$

which shows that $\hat{\theta}_{AVG}(X_1, \dots, X_n)$ is unbiased.

(b) Compute $\text{Var} \left(\hat{\theta}_{AVG}(X_1, \dots, X_n) \right)$.

- **Solution:** The variance of a $\text{Uniform}(0, \theta)$ random variable is

$$\text{Var}(X_i) = \frac{\theta^2}{12}.$$

Since $\bar{X}_n$ has variance

$$\text{Var}(\bar{X}_n) = \frac{\theta^2}{12n},$$

it follows that

$$\text{Var}(2\bar{X}_n) = 4\text{Var}(\bar{X}_n) = \frac{4\theta^2}{12n} = \frac{\theta^2}{3n}.$$

- (c) Let $X_{(n)} = \max_{i=1, \dots, n} X_i$ denote the largest of all X_i 's. Show that the cumulative distribution function of $X_{(n)}$ is given by:

$$F_{(n)}(x; \theta) = \begin{cases} 0 & \text{if } x < 0 \\ \left(\frac{x}{\theta}\right)^n & \text{if } x \in [0, \theta] \\ 1 & \text{if } x > \theta. \end{cases}$$

- **Solution:** For $X_i \sim \text{Uniform}(0, \theta)$, the cdf of each X_i is

$$F_X(x) = \frac{x}{\theta}, \quad 0 \leq x \leq \theta.$$

Since the X_i are i.i.d., the cdf of $X_{(n)} = \max\{X_1, \dots, X_n\}$ is

$$F_{X_{(n)}}(x) = (F_X(x))^n = \left(\frac{x}{\theta}\right)^n \quad \text{for } x \in [0, \theta],$$

with $F_{X_{(n)}}(x) = 0$ for $x < 0$ and $F_{X_{(n)}}(x) = 1$ for $x > \theta$.

- (d) Show that the maximum likelihood estimator for θ is given by $\hat{\theta}_{MLE}(X_1, \dots, X_n) = X_{(n)}$.

Hint: In this particular example, the likelihood function will not be differentiable for all values of θ – and so you won't be able to use our usual approach to find its argmax. Instead, study the likelihood function itself – where is it equal to zero and why? Where is it not differentiable?

- **Solution:** The likelihood function is

$$\mathcal{L}(\theta; X_1, \dots, X_n) = \prod_{i=1}^n \frac{1}{\theta} \mathbb{I}_{\{0 \leq X_i \leq \theta\}} = \frac{1}{\theta^n} \mathbb{I}_{\{X_{(n)} \leq \theta\}},$$

where $X_{(n)} = \max\{X_1, \dots, X_n\}$. Notice that if $\theta < X_{(n)}$, then the likelihood is zero, while for $\theta \geq X_{(n)}$, $\frac{1}{\theta^n}$ is a decreasing function of θ . Therefore, the maximum is achieved when $\theta = X_{(n)}$, i.e.,

$$\hat{\theta}_{MLE}(X_1, \dots, X_n) = X_{(n)}.$$

- (e) Compute $\mathbb{E}[\hat{\theta}_{MLE}(X_1, \dots, X_n)]$. Is this estimator unbiased? If not, how could you construct a different unbiased estimator from $\hat{\theta}_{MLE}(X_1, \dots, X_n)$?

• **Solution:** For $X_i \sim \text{Uniform}(0, \theta)$, it is known that

$$\frac{X_{(n)}}{\theta} \sim \text{Beta}(n, 1),$$

so that

$$\mathbb{E}\left[\frac{X_{(n)}}{\theta}\right] = \frac{n}{n+1} \implies \mathbb{E}[X_{(n)}] = \frac{n}{n+1}\theta.$$

Since $\mathbb{E}[X_{(n)}] \neq \theta$, the estimator $\hat{\theta}_{MLE}(X_1, \dots, X_n) = X_{(n)}$ is biased. An unbiased estimator can be constructed by scaling $X_{(n)}$:

$$\tilde{\theta} = \frac{n+1}{n}X_{(n)},$$

which satisfies

$$\mathbb{E}[\tilde{\theta}] = \frac{n+1}{n}\mathbb{E}[X_{(n)}] = \theta.$$

- (f) Compute $\text{Var}\left(\hat{\theta}_{MLE}(X_1, \dots, X_n)\right)$.

• **Solution:** Again using the fact that $X_{(n)}/\theta \sim \text{Beta}(n, 1)$, the variance of a $\text{Beta}(n, 1)$ random variable is

$$\text{Var}\left(\frac{X_{(n)}}{\theta}\right) = \frac{n \cdot 1}{(n+1)^2(n+2)}.$$

Thus,

$$\text{Var}(X_{(n)}) = \theta^2 \frac{n}{(n+1)^2(n+2)}.$$